

SE-Overview of It Industry

Name: Goswami Mansi

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans. HTTP (HyperText Transfer Protocol)

Used for transferring web pages and data between a browser and a web server.

- **HTTPS (HyperText Transfer Protocol Secure)**

A secure version of HTTP that encrypts data for safe communication.

- **FTP (File Transfer Protocol)**

Used to transfer files between computers over a network.

- **SMTP (Simple Mail Transfer Protocol)**

Used for sending emails from one mail server to another.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans. IPv4 uses 32-bit addresses and supports about 4.3 billion unique addresses.

- IPv6 uses 128-bit addresses and supports a much larger number of unique addresses.

3. Run nslookup google.com on your system and write down the IP address you get.

Ans. IP address: 142.251.142.206

Note: The IP address may be different on your system.

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans. 200 OK – Request was successful.

- **301 Moved Permanently** – Resource has been permanently moved to a new URL.
- **404 Not Found** – Requested page was not found.
- **500 Internal Server Error** – Server encountered an error.
- **403 Forbidden** – Access to the resource is denied.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans. Root (.) → Top level of DNS hierarchy.

- **TLD (com)** → Top-Level Domain.
- **Domain (google)** → Registered domain name.
- **Subdomain (www)** → Subdomain of google.com.

Example:

www.google.com

- www = Subdomain
- google = Domain
- com = TLD
- . (implicit) = Root

